

Rongfeng Cheng | Medical AI Research

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Research Interests

Trustworthy medical AI, ECG and physiological-signal learning, model auditing and external validation, multimodal representation learning, and clinical NLP.

Education

South China University of Technology (SCUT)

Guangzhou, China

Bachelor of Management in Big Data Management and Application

2023–2027

GPA: 3.46/4.0.

Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

Research Experience

Institute of Healthcare Artificial Intelligence Application, Guangdong Second Provincial General Hospital

Evidence-Use Auditing for Deep ECG Models

Jan 2026–Present

Advisors: Haobin Zhang and Weibin Cheng.

- Solely led study design, implementation, experiments, analysis, and manuscript preparation; developed a matched-control perturbation audit spanning clinically defined evidence units, operator comparisons, and frozen-weight external transfer.
- Evaluated five diagnoses and three raw-signal models on CODE-15% and PTB-XL; the audit classified 6 of 15 model-diagnosis pairs as evidence-supported and showed that matched controls or operator choice can overturn target-only conclusions. See Manuscript [1].

DLEF: Dataset-Label Evidence Fingerprints

Jun 2026–Present

Advisors: Haobin Zhang and Weibin Cheng.

- Solely led study design, implementation, experiments, analysis, and manuscript preparation; proposed directional evidence fingerprints for characterizing dataset-label transfer risk.
- Across CODE-15%, PTB-XL, and Chapman-Shaoxing, DLEF distance was associated with label-level external AUROC degradation (Spearman $r = 0.843$ across 108 points; $r = 0.892$ after source-target-label aggregation). See Manuscript [2].

PathCLIP: Evidence-Aware ECG–Text Representation Learning

Jul 2026–Present

Advisors: Haobin Zhang and Weibin Cheng.

- Leading experimental implementation for ECG–text alignment and zero-shot recognition, including matched-count, permuted-hierarchy, residual-only, and held-out-label controls that separate knowledge effects from added supervision or capacity.

Manuscripts & Publications

[1]: R. Cheng et al. “A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification.” Under review, *IEEE JBHI*. First author.

[2]: R. Cheng et al. “DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification.” Under review, *IEEE BIBM 2026*. First author.

[3]: J. He, S. Qin, H. Zou, F. Jing, W. Liu, C. Li, Q. Wang, Z. Hu, K. Song, Z. Xu, Q. Luo, Y. Cheng, R. Cheng, H. Zhang, and W. Cheng. “CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation.” *iScience*, 2026. doi:10.1016/j.isci.2026.115073.

[4]: R. Wang, F. Jing, R. Cheng, Y. Cheng, W. Cheng, H. Zou, H. Ren, and W. Jin. “Artificial Intelligence Applications in X-ray Coronary Angiography: A Comprehensive Review of Image Analysis Techniques.” *HNNDL 2026*. doi:10.1145/3795892.3795917.

[5]: Y. Fang, F. Jing, R. Cheng, Y. Cheng, W. Cheng, H. Ren, W. Cai, and W. Jin. “Artificial Intelligence for Echocardiographic Assessment of Pulmonary Hypertension and Right Heart Function: Advances in Intelligent Diagnosis and Risk Stratification.” *HNNDL 2026*. doi:10.1145/3795892.3795918.

Clinical Research & Engineering Experience

Institute of Healthcare Artificial Intelligence Application, Guangdong Second Provincial General Hospital
Research Intern Nov 2024–Present

- Solely developed a standardized-patient Q&A pipeline from approximately 8,000 cardiovascular case PDFs, covering LLM-based information extraction, entity disambiguation, Neo4j knowledge representation, and hybrid retrieval.
- Led full-stack and clinical-workflow development for the five-person Dingbei Meimei acne and facial-health assessment team; supported its initial release when the hospital's Refractory Acne Diagnosis and Treatment Center opened in July 2026.

Additional Research Projects

Project Lead, SCUT Student Research Program Guangzhou, China
Myocardial Ischemia Knowledge Graph 2025
Advisor: Xunde Dong. Led a five-student interdisciplinary team in building a domain knowledge graph with collaborative ontology management, XGBoost-based implicit reasoning, and graph-mapped interpretability analysis; completed the program successfully.

Independent Research
IHD Trajectory Similarity Retrieval Nov–Dec 2025
Developed trajectory retrieval and post-cutoff validation for ischemic heart disease ICU stays in MIMIC-IV using time-windowed clinical variables and precomputed k-NN similarity graphs.

Patents & Software IP

Patent: Method, System, and Electronic Device for Medical Question-Answering Output Assurance – First Inventor; application #202610582199.6; filed Apr 2026; preliminary examination passed.

Software IP: Knowledge Graph Ontology Intelligent Management System V1.0 (#2025SR2454864); LLM-Based Cardiovascular Proactive Health Management Intelligent System V1.0 (#2025SR2080673).

Honors

- First Prize, 2025 Challenge Cup (Jinan University Campus).
- Second Prize, 2025 9th Guangzhou Mathematical Modeling University League.
- Third Prize, 2025 National Mathematical Modeling Contest – Guangdong Division.

Skills

Modeling: PyTorch, XGBoost, representation learning, multi-modal learning	Biomedical Data: 12-lead ECG, MIMIC-IV / EHR cohorts, clinical NLP, knowledge graphs
Evaluation: Perturbation analysis, matched controls, cross-dataset evaluation	Engineering: Python, TypeScript, SQL, Next.js, Flask, Neo4j, Git, Docker

Languages

Chinese (native) Cantonese (native) English: IELTS 6.5, CET-6 528